

Arden Tsang

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EDUCATION

University of Cambridge

MASt in Astrophysics

Expected graduation: Jun 2027

Cambridge, UK

- **First Class Honours.**

- QFT[x], GR[x], Category Theory[x], Field Theory in Cosmology[x], Gravitational Waves & Numerical Relativity[x]
- Unassessed: Solitons, Instantons & Geometry; Gauge/Gravity Duality; Standard Model; String Theory; Quantum Information & Gravity

King's College London (KCL)

BSc Physics with Theoretical Physics

Expected graduation: Jun 2026

London, UK

- **First Class Honours** every year, Ranked **Top 5%** of year.
- **King's Research Award.**
- QM[81], EM[83], Math in Physics[79] & Theoretical Physics[77], Math & Compute[76], Symmetry in Physics[76].

RESEARCH & EXPERIENCE

CCP5 RESEARCH FELLOW; THEORETICAL & COMPUTATIONAL MATERIALS PHYSICS

Dec 2025 – Present

KCL De Tomas Group

London, UK

- **Awarded EPSRC-funded CCP5** (Condensed Matter Simulation Project) Bursary for full-time summer extension.
- Developed and characterised a hybrid PSO–DFT (GFN2-xTB/PBE+VV10) workflow for carbon-nanocluster structure prediction; benchmarked inertias, recovered C_{3-10} isomers and converged a C_{20} cage, fullerene precursor.
- Built HPC pipelines in Bash for GPU-accelerated optimiser tuning; authored READMEs for reproducibility. [[Git](#)]
- Curated benchmark datasets to validate against GAP-20/AIRSS, bridging ab initio accuracy with ML-IP scalability.
- Extending to compare simulated IR spectra against astrophysical molecular cloud observations. Supervised by Dr Carla De Tomas, Senior Lecturer in Net Zero.

PARTICLE PHYSICS & MACHINE LEARNING RESEARCH ASSISTANT

Oct 2025 – Present

KCL Experimental Particle and Astroparticle group (EPAP)

London, UK

- **Co-authored paper:** “Physics at the Edge: Benchmarking Quantisation Techniques and the EdgeTPU for Neutrino Interaction Recognition”. ([arXiv:2603.24607](https://arxiv.org/abs/2603.24607), awaiting submission to EPJC)
- Presented poster at the AI for Science Conference, Alan Turing Institute, March 2026.
- Architecting ground-up, lightweight, symmetry-preserving models for EdgeAI topology classification.
- Trained and quantized CNNs (PTQ & QAT) with GENIE generated LArTPC neutrino datasets to support possible low latency, sustainable AI fast triggers. Supervised by Dr Stefano Vergani, Postdoctoral Research Associate.

PHOTOVOLTAICS UNDERGRADUATE RESEARCH ASSISTANT

Sep 2025 – Present

KCL Functional Materials & Devices Laboratory

London, UK

- Integrated CAD: Fusion360 & Inkscape with V4.75 CO₂ laser to nanostructure perovskite for spectral filtering.
- Investigated dichroic filters, perovskite for agrivoltaics; Supervised by Dr Jay Patel, Functional Photonics Lecturer.

ARTIFICIAL INTELLIGENCE UNDERGRADUATE SUMMER RESEARCH FELLOW

Jul 2025 – Sep 2025

HKUST Industrial Engineering and Decision Analytics (IEDA)

Hong Kong

- Designed and programmed scheduling algorithms for NPU architectures to optimize inference speed.
- Learned transformer theory & curated datasets to fine-tune LLMs with mlx to test physical derivation recovery.
- Interviewed published researchers on AI safety. Supervised by Dr Jiheng Zhang, Professor of IEDA. & Mathematics.

QUANTUM CHEMISTRY KING'S UNDERGRADUATE RESEARCH FELLOW (KURF)

Jun 2025 – Jul 2025

KCL Physical Chemistry (Sanz) Group

London, UK

- Successful identification of the lowest-energy isomer of limonene–(SO₂)₂ cluster by rotational spectroscopy.
- Employed Quantum Chemistry suite: DFT on the HPC via CLI; Spectral assignment, & NCI, NBO Chemical analysis.
- Awarded paid research fellowship; Supervised by Dr Maria Eugenia Sanz, Reader in Physical Chemistry.

ACTUARIAL INTERN

Jul 2024 – Sep 2024

AIA

Hong Kong

- Studied credit risk algorithms & created market projections, executed data analysis & crafted quarterly report.
- Documented linear algebra legacy code in Python and VBA from Excel to senior colleagues.

PUBLICATIONS, POSTERS & CONFERENCES

- Vergani, S.*, **Tsang, A.**, et al. Physics at the Edge: Benchmarking Quantisation Techniques and the Edge TPU for Neutrino Interaction Recognition. **AI for Science Conference, Alan Turing Institute**, London, UK. Mar 2026
- **Tsang, A.**, Kuchmina, S., de Tomas, C., et al. Simulating Carbon Nanoclusters using ML-IP & Particle Swarm Optimisation. **London Centre of Nanotechnology Symposium**; BSc Thesis Poster Session, London, UK, Apr 2026
- **Tsang, A.**, Sanz, M. E. Atmospheric Clusters: Structure and Interactions. Academic poster submitted for the King's

POSITIONS OF RESPONSIBILITY

Cosmology Hackathon, KCL Department of Physics

(London, UK) Jun 2026

- Built a ML pipeline to infer galaxy redshifts from 5-band SDSS photometry, combining gradient-boosted trees and an MLP ensemble; RMSE $\approx$ [0.052].

Staff Writer, King's International Security Journal (KISJ)

(London, UK) Sep 2025 – Present

- Sole author on AI's authoritarian tendencies, misalignment, collapse of information scarcity & macro-economy

Engineering, Philosophy Lead, KCL Rocketry Society

(London, UK) Sep 2024 – Feb 2026

- Showcased technical exhibition & facilitated physical rocket construction utilizing Fusion360 for modeling.

Elected Head, SMUS Physics Club

(Victoria, BC, Canada) Sep 2022 – Jun 2023

- Executed 15+ experiments, tested hypotheses, argued methods & results for theory; Hosted lectures on QM.

Co-Founder & Head, SMUS Engineering Club

(Victoria, BC, Canada) Sep 2021 – Jun 2022

- Recruited 80+; Lectured; Directed bridge competition; Taught Inkscape to design & compete with printed car.

Physics Competitions; [IOP/IAPS], [CAP/IQC]

(Swansea, UK); (Victoria, BC, Canada) Sep 2021 – Feb 2025

- **PLANCKS**: Winner of promotional event, competed & networked across UK universities in theory.
- **CaYPT**: Led team of 5 to design experiments, debated methods, results, & theories with schools across Canada.

STEM Tutor; Various Institutions

(Victoria, BC, Canada); (London, UK) Sep 2021 – Present

- Instructed 15+ students in AP & IB Physics, AP Calc, Chem, Stat; across English & Cantonese, remote & in-person

TECHNICAL SKILLS

- **Languages**: Native English, Native Cantonese, Fluent Mandarin, Limited Working Proficiency Spanish.
- **Programming & Computation**: Python (PySCF, gpu4pySCF, PyTorch, TensorFlow, Keras, MLX, SciPy, Pandas, Numpy, Matplotlib), MATLAB; Java; VBA; Swift;
- **HPC & Systems**: Linux/Unix CLI (MobaXterm, MacOS, WindowOS), Schedulers, GPU-accelerated workflows.
- **Software & Tools**: $\LaTeX$; Git; Quantum Chemistry: PGOPHER, Pickett, Gaussian, Chimera, Multiwfn; CAD: Fusion360, Inkscape, Prusa; OVITO, NetworX; Video editing/photoshop; Sketchup; Microsoft Office Suite.
- **Machine Learning & Numerical Methods**: LLM fine-tuning (MLX); CNNs; Quantization: (Post-Training-Quantization, Quantization-Aware-Training), Stochastic Optimisation (PSO, Simulated Annealing); ODEs/PDEs (Mckenna's model, Taylor's, Tacoma); Packing (NPU); QM, MRI and ultrasound theory & visualisation.
- **Mathematical Physics**: Lie Algebras & Representation Theory; Analytical Mechanics; Quantum, Relativistic & Statistical Mechanics; Tensor Calculus; Axiomatic Set Theory.
- **Experimental Physics**: Optics & Photonics: Interferometry, Laser alignment, Spectroscopy, Polarimetry. Instrumentation: CO₂ lasers, Optical bench, Oscilloscopes, Helmholtz coils, NMR Spectrometers, Circuit design.
- **Leadership** abilities from society administrations & diverse collaborations across international environments.